

PROJECT PROPOSAL

ArthoBachao AI

An AI-Powered, Bilingual Personal Finance Coach for Bangladesh

Item	Details
Project Title	ArthoBachao AI: An AI-Powered, Bilingual Personal Finance Coach for Bangladesh
Project Type	Full-stack, AI-integrated web application (working prototype to pilot-ready product)
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1. Executive Summary

ArthoBachao AI is a web-based personal financial health and savings assistant designed specifically for people in Bangladesh. It turns everyday transactions across bKash, Nagad and local bank accounts into clear spending analysis, a transparent financial health score, a 30-day cash-flow forecast, savings-goal tracking, and an AI coach that answers money questions in natural Bangla or English.

Most budgeting tools are built for Western banking habits, priced in dollars and available only in English. ArthoBachao AI is built around Taka (₳), mobile financial services, cash-out fees, Dhaka-style living costs and conversational Bangla, so that advice feels relevant and trustworthy. Every answer from the coach is grounded in the user's own ledger data and explains its reasoning.

A working prototype (MVP) already demonstrates the core experience. This proposal covers hardening the system, adding real data input (statement import), building smarter planning tools, and piloting the product with 30–50 real users over a 12-week plan.

2. Background and Problem Statement

Mobile financial services such as bKash and Nagad, together with digital banking, have made day-to-day payments in Bangladesh fast and convenient. Convenience, however, has not translated into financial awareness. Many individuals, particularly young professionals and students in urban areas, face the following problems:

1. **Month-end cash pressure.** People run short of money before month-end and cannot tell exactly why.
2. **Fragmented money trail.** Spending is spread across MFS wallets, bank cards, cash and transfers, so no single view shows where money goes.
3. **Invisible small costs.** Recurring cash-out charges, bank fees and delivery fees add up without being noticed.
4. **No local, Bangla-first tool.** Existing international apps use foreign currency, English-only interfaces, and ignore local categories and merchants.
5. **Generic, unexplained advice.** Generic chatbots give generic answers, and personal advice is otherwise available only through paid advisors. Users need guidance that cites their own numbers.
6. **Weak goal planning.** Users set savings goals but have no clear picture of the monthly amount required or the projected completion date.

There is therefore a clear need for an accessible, explainable and locally relevant tool that turns raw financial data into practical, personalised guidance.

3. Objectives

3.1 General Objective

To design, harden and pilot an AI-powered personal finance assistant that improves the financial awareness and savings behaviour of users in Bangladesh.

3.2 Specific Objectives

1. Provide secure user registration and authentication.
2. Give users one clear picture of income, spending, savings rate and balances in Taka.
3. Record and categorise transactions, detect spending patterns, month-on-month changes and unusual transactions, and explain them in plain language.
4. Compute a transparent Financial Health Score from saving consistency, spending control, cash-flow stability and goal progress.
5. Forecast cash flow for the next 30 days and warn users of pressure days before money runs out.
6. Help users set, track and reach savings goals with realistic monthly targets and projected completion dates.
7. Offer a conversational AI coach in Bangla and English, grounded strictly in the user's own ledger data.
8. Keep the product reliable: it must keep working even when the AI service or the database is unavailable.

4. Target Users

1. Salaried professionals and young earners in Dhaka and other cities who use bKash, Nagad and bank cards daily.
2. University students and fresh graduates building their first savings habit.
3. Households planning for goals such as an emergency fund, education, marriage or a major purchase.
4. Users who prefer Bangla and find existing finance apps hard to follow.

5. Scope of the Project

5.1 In Scope

1. Responsive web application with a React front end and Node.js/Express back end, backed by MongoDB.
2. REST APIs for authentication, users, transactions and goals.
3. AI Coach endpoint integrated with the Google Gemini API.
4. Bilingual (Bangla and English) interaction and BDT-based reporting.
5. Statement import (CSV/PDF), manual entry and manual category correction.

6. Budgets, bill reminders and scenario simulation.
7. A pilot with 30–50 users.

5.2 Out of Scope (Current Phase)

1. Direct integration with bank or mobile-wallet APIs (bKash, Nagad, etc.); to be pursued later through official partnerships.
2. Executing payments or transfers on behalf of users.
3. Investment, tax-filing or insurance advisory services.
4. Native mobile applications (a PWA / mobile layout is in scope instead).

6. Proposed Solution and Key Features

Feature	What it does
Secure accounts	Registration and login with hashed passwords (bcrypt) and token-based sessions (JWT); auth middleware protects all user data routes.
Dashboard	Monthly overview of income, expenses, net savings, savings rate, daily average spend and linked account balances.
Transactions & spending analysis	Add, view and categorise income and expenses. Breaks spending into categories (Food & Groceries, Shopping, Utilities, Transportation, Cash-out & Bank Fees, Entertainment), shows share and month-on-month change, and splits essential vs discretionary spending.
Anomaly & fee-leak insights	Flags unusual transactions and rising fees with severity, confidence and step-by-step recommended actions, including estimated savings.
Financial Health Score	A transparent 0–100 score built from four pillars (Saving Consistency, Spending Control, Cash-Flow Stability, Goal Progress) with a written explanation.
30-day cash-flow forecast	Projects the daily balance curve, upcoming bills and expected inflows, highlights pressure days, and recommends a safety buffer.
Savings goals	Target, saved amount, deadline, milestones, required monthly saving and monthly gap. Shows On Track / At Risk status and projected completion date; transfers between vaults are supported.
AI financial coach	Chat assistant powered by Google Gemini. Answers questions such as “Why do I run short before month-end?”, “Which category should I reduce?” and “How can I save more?” using the user’s real figures.
Monthly report	Printable-style monthly summary of the user’s financial position.
Bilingual interface	One-click switch between English and বাংলা across the whole app; all amounts in Taka (৳).

7. System Architecture and Technology Stack

The application follows a three-tier architecture: a single-page React client, an Express API layer, and a MongoDB data layer. The Gemini API is an external intelligence service called only from the server, so the API key is never exposed to the browser.

Layer	Technology
Front end	React 19, TypeScript, Vite, Tailwind CSS 4, Motion (animations), Lucide icons; Plus Jakarta Sans and Noto Sans Bengali fonts
Back end	Node.js with Express, written in TypeScript (run with tsx)
Database	MongoDB with Mongoose (User, Transaction and SavingsGoal models)
AI service	Google Gemini via the @google/genai SDK (Flash-family models), server-side only
Security	JWT tokens, bcryptjs password hashing, auth middleware, secrets in environment variables (dotenv)
Testing	Script-based API tests for authentication, Bangla and edge-case prompts, fallback behaviour and Gemini validation; to be broadened in Phase 1

7.1 API Overview

Endpoint	Purpose
/api/auth	Registration and login
/api/users	User profile and user-scoped transactions
/api (transactions, goals)	Create and retrieve transactions and savings goals
/api/ai/coach (POST)	Receives the user prompt, language flag and financial context; returns the coach response and its source
/api/health (GET)	Reports server status, database connectivity and AI-key availability

8. AI Methodology: Accurate, Explainable and Reliable

8.1 Grounded Prompting

For every coach request, the server builds a system instruction from the user's real ledger: income, spending, net savings, savings rate, available balance, category breakdown with month-on-month change, goal status, health-score factors and the cash-flow forecast. The model is told to cite exact figures, explain causes, give Bangladesh-specific suggestions (for example mobile-wallet cash-out fees or food-delivery spending), use Taka only, and never invent numbers.

8.2 Calculations in Code, Language in the Model

All numbers (totals, scores, forecasts, goal pace) are computed by deterministic application code. The model only explains and advises on figures it is given, which keeps advice consistent with the data and limits hallucination.

8.3 Layered Resilience

1. **Model fallback.** The server tries an ordered list of approved Gemini models and retries transient errors (HTTP 429/503) once before moving to the next model.
2. **Rule-based fallback.** If every AI call fails or no key is configured, a deterministic engine answers common questions (month-end shortage, which category to cut, how to save more, top spending driver, goal status, next-month balance) in both languages from the same user data.
3. **Database fallback.** The database connection is established safely at start-up and never crashes the server; if MongoDB is unavailable the app runs in a standalone mode and /api/health reports the state.
4. **Transparency.** Each coach response states its source (Gemini or rule-based fallback), so behaviour is always visible to the user.

9. Current Status

The project is a working prototype. The table separates what exists today from what this proposal will deliver.

Status	Items
Completed	Full user interface with all main screens; English and Bangla support; spending, health-score and cash-flow calculations; savings goals; AI coach with Gemini and fallbacks; REST API; MongoDB models; registration and login; demo data for Taka-based transactions.
In progress / to harden	Run all calculations on live stored data; stronger input validation; remove default development secrets and enforce environment-based configuration; server-side rate limiting; broader automated tests.
Planned	Statement import (CSV/PDF) with AI categorisation; budgets and bill reminders; scenario simulation; push notifications for pressure days; PWA / mobile layout; seasonal forecasting (e.g. Eid); data export; later, partner or API-based MFS and bank connections.

10. Development Methodology and Timeline

The work follows an iterative (Agile-style) approach with short cycles of design, implementation, testing and review, building on the existing prototype. The 12-week plan can be adjusted to the institutional calendar.

Weeks	Activity	Deliverable
1 to 3	Phase 1, stabilise the core: connect every screen to stored data, security hardening, input validation, broader tests	Stable core, expanded test suite, deployment pipeline (HTTPS)
4 to 6	Phase 2, real data input: statement import (CSV/PDF), automatic AI categorisation, duplicate detection, manual category correction	Working statement import
7 to 9	Phase 3, smarter planning: budgets, bill reminders, scenario simulation (“what if I cut food delivery by 20%?”), improved forecast	Budget and simulation modules
10 to 12	Phase 4, launch and feedback: PWA / mobile layout, pilot with 30–50 users, feedback collection and fixes, final documentation	Pilot results, test report, final report and presentation

11. Expected Outcomes and Success Measures

Targets below are proposed and can be refined with the supervisor before the pilot begins.

Measure	How it is measured	Proposed target
Speed to insight	Usability test: time for a new user to see their full monthly picture	Under 1 minute
Coach accuracy	Test prompts in Bangla and English; coach figures compared with the ledger	At least 95% of answers match ledger figures
Coach availability	Requests answered while Gemini is simulated as unavailable	Every request receives an answer
Goal planning	Share of users with a goal who see a monthly target and completion date	100%
Categorisation quality	Share of imported transactions corrected manually by pilot users	Tracked; reduced over the pilot
Behaviour change	Pilot survey and data: month-end shortfalls and savings rate, before vs during pilot	Fewer shortfalls and a higher savings rate

Beyond these measures, the project contributes a locally contextualised reference design for AI-assisted personal finance in Bangladesh, and lowers the barrier to financial literacy by giving explainable advice in the user’s own language.

12. Risks and Mitigation

Risk	Impact	Mitigation
Privacy of financial data	Loss of user trust	Read-only data handling; hashed passwords; token authentication; HTTPS; secrets in environment variables; no storage of PINs or OTPs; minimum data collection; clear privacy notice.
AI gives wrong numbers	Misleading guidance	Grounded prompts with “cite exact figures” rules; calculations done in code, not by the model; low temperature; clear disclaimer.
AI downtime, limits or cost	Coach unavailable	Multi-model fallback, retry logic, rule-based answers, server-side rate limiting.
Database unavailable	Data not persisted	Safe connection handling, standalone mode and health monitoring.
Difficulty getting bank/MFS data	Limited automation	Start with statement import and manual entry; pursue official partnerships and APIs later.
Low trust or adoption	Limited impact	Every score and recommendation comes with an explanation; bilingual, simple UI; positioned as a coaching tool, not licensed financial advice.

13. Ethical, Security and Privacy Considerations

Financial data is highly sensitive. The system sends the AI service only the information needed for each answer, stores credentials only in hashed form, keeps secret keys in environment variables excluded from version control, and presents advice as educational guidance rather than professional financial, legal or investment advice. Pilot participants will give informed consent, and compliance with applicable data-protection requirements will be reviewed before any public release.

14. Future Enhancements

1. Automatic transaction import via SMS parsing or authorised bank and wallet APIs.
2. Push notifications for pressure days and unusual spending.
3. Receipt scanning and voice input in Bangla.
4. Native mobile application and offline support.
5. Audit logging and a full production-grade automated test suite.

15. Conclusion

ArthoBachao AI combines practical financial analytics with a Bangla-friendly AI coach to help ordinary people in Bangladesh understand where their money goes and build better habits. The prototype already demonstrates the core experience; the work proposed here will turn it into a dependable product that uses real user data, protects privacy, and is validated with real users.

We respectfully request approval to proceed with its development and pilot evaluation as outlined in this proposal.

Submitted by:

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[Signature and Date]